BGIN Block#12 Meeting Report

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Financial Stability Risk within Decentralized Financial System

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BGIN Block#12

Financial Stability Risk within Decentralized Financial System

Date: Monday, March 3, 2025

Location: Tokyo, Japan

Duration: Approx. 90 minutes

Summary

- Purpose of the Session:
    - Introduce the concept of liquid staking, restaking, and liquid restaking to traditional finance experts, and highlight financial stability risks associated with these leveraged staking mechanisms in decentralized finance (DeFi).
    - Understand how these risks could propagate from DeFi to traditional financial systems and identify the next steps for further research.

1. Foundational Concepts Clarified:

- Explanation of Liquid staking (LSTs):
    - Traditional staking on Ethereum requires 32 ETH, but protocols like Lido allow users to stake smaller amounts and receive a liquid staking token (LST) such as stETH in return.
    - Liquid staking protocols aggregate user deposits and delegate them to Ethereum validators. Users retain liquidity via LSTs, which can be traded or used in DeFi.
    - Lido and EigenLayer are separate. While Lido issues LSTs like stETH, EigenLayer introduces restaking. Users can restake native ETH or stETH on EigenLayer to secure additional services (AVSs).
- Explanation of Restaking:
    - Restaking allows staked ETH (which is normally idle) to secure other protocols, increasing capital efficiency.
    - Validators (operators) give control of their withdrawal credentials to EigenLayer, enabling slashing enforcement across multiple protocols (AVSs).
    - Operators register with AVSs (Actively Validated Services) and agree to their slashing conditions. AVSs get economic security from staked ETH. Misbehavior (e.g., providing bad oracle data) can result in slashing.

- Risk associated with restaking includes:
  - Economic Security Risks: If operators are slashed by one AVS, they lose ETH staked on Ethereum, which could impact multiple protocols they support.
  - Complex Slashing Conditions: Currently, most AVSs don't have clear slashing rules, making it hard to quantify or anticipate risk.
  - Reward Misalignment: Operators may choose AVSs based on token incentives rather than technical fit or safety, creating potential systemic risks.

- Liquid restaking:
  - Even restaked assets can be tokenized, creating *liquid restaking tokens (LRTs)* that can be traded or reused, compounding complexity and leverage.

2. Economic Security Model of Restaking:

- EigenLayer mediates slashing: if an operator misbehaves on any AVS, EigenLayer may *slash their ETH*, preventing withdrawal or reducing payout. This mechanism enables AVSs to *enforce accountability* without having to bootstrap their own security. However, slashing conditions are not fully implemented yet, so current analysis remains partly theoretical.

3. Risks Highlighted in Restaking Model:

- Cascading risks:
  - A slashing event in one AVS may cause loss of validator funds, which affects their ability to secure other AVSs. The *entire economic security structure* can be undermined through domino-like failures.

- Network structure:
  - Operators tend to *subscribe to multiple AVSs* to maximize rewards, creating a *bipartite graph*. Sparse graphs (low AVS participation) reduce security, while dense graphs (high participation) increase slashing exposure.

- A participant presented a *simulation* showing how these cascading failures can propagate in the restaking network based on actual data. An operator failure can lead to slashing in multiple AVSs, which in turn affects other operators, creating a chain reaction.

- Users can get LSTs and liquid restaking tokens (LRTs), which can be used as collateral in DeFi.
  LSTs and LRTs can be reused in lending protocols, creating leverage. If token prices fall

(e.g., due to slashing), mass liquidations may occur. LSTs may lose peg, triggering liquidation cascades and affecting the broader DeFi ecosystem.

4. Comparison to Traditional Finance:

- Operators were likened to *brokers*, and EigenLayer to a *clearing house* or *insurance system*. One analogy: operators take credit risk like brokers do, while EigenLayer guarantees enforcement like a clearing house. However, restaking adds complexity via *token reuse* and *stacked leverage*, similar to *rehypothecation* in TradFi.

- This sparked discussion on whether these analogies can help *quantify risk* and how *portfolio theory* (e.g. Markowitz optimization) may apply to operator decisions. A participant noted the difficulty in quantifying both risk and reward due to lack of real-world lashing events and off-chain governance.

5. Slashing Correlation and Systemic Risk:

- Debate emerged on *how correlated slashing events* across AVSs could be:
    - Some argued they are *idiosyncratic* since slashing is tied to specific misbehavior (e.g., wrong price feed in an oracle AVS). Others warned that *infrastructure failure* (e.g., AWS downtime affecting many validators) could lead to *mass slashing*.

- Ethereum's slashing rules are *highly specific* (e.g., double-signing) and downtime doesn't typically result in slashing—mitigating some correlation concerns. Slashing is designed to be narrowly targeted in general, reducing likelihood of contagion.

- Still, the possibility of unexpected correlation spikes (like during the 2008 housing crash) was raised as a valid risk research area. Some voiced concern that unknown risks could lead to correlated slashing events, analogous to the 2008 housing crisis.

6. Propagation to Traditional Finance:

- Concern raised: if only DeFi is impacted, regulators may not act—but *DeFi risks can propagate* to TradFi. For example, ETH ETFs or staking services offered by TradFi could expose investors to *underlying risks* they don't understand. Collapse of LST prices could affect lending protocols, which could indirectly affect centralized entities with DeFi exposure.

- Agreed on the need to map these interconnections to quantify potential systemic risk spillovers.

- A study was cited estimating that leveraged staking strategies have an *average multiplier of 3.5x* and APR of ~6% (2020–2023). Discussion explored whether APRs from staking were *correlated with interest rate environments*:
    - So far, ETH staking yields appear *independent* from macro rates—driven more by *validator count* and *network activity*. APR growth seems to have plateaued at ~3%, slowing new staking inflows.

7. Preventative Measures & Ethereum Foundation's Role:

- Question raised: are *any preventative controls* in place in DeFi or Ethereum to limit LSD/LRT risk propagation? Answer: measures exist *within specific protocols*, but no *ecosystem-wide mitigation framework* yet. Preventive design (e.g., caps on collateralization) would have to be protocol-specific. Ethereum Foundation hasn't taken official action on EigenLayer yet.

- The Ethereum Foundation reportedly considered protocol changes (e.g., disallowing smart contracts as withdrawal addresses), but these weren't implemented.
    - Ethereum governance may step in if slashing from AVSs threatens ETH consensus.
    - There were internal discussions (e.g., Justin Drake's talk) on whether to prevent smart contracts from being withdrawal addresses—a way to prevent EigenLayer-like restaking.
    - One concern was that *Ethereum may avoid restrictive action* for fear of driving users to other PoS chains like Solana or BNB Chain.

8. Next Steps:

- There was a call for deeper research, possibly by BGIN or affiliated universities, to:
    - Quantify risks in leveraged staking and restaking.
    - Identify potential amplification mechanisms.
    - Track the interlinkages between DeFi and TradFi, especially through derivative tokens and staking-based financial products.

- The conversation ended with recognition of the importance of cross-disciplinary collaboration between DeFi researchers, traditional finance experts, and regulators to build a comprehensive understanding of the risks and rewards of these evolving technologies.